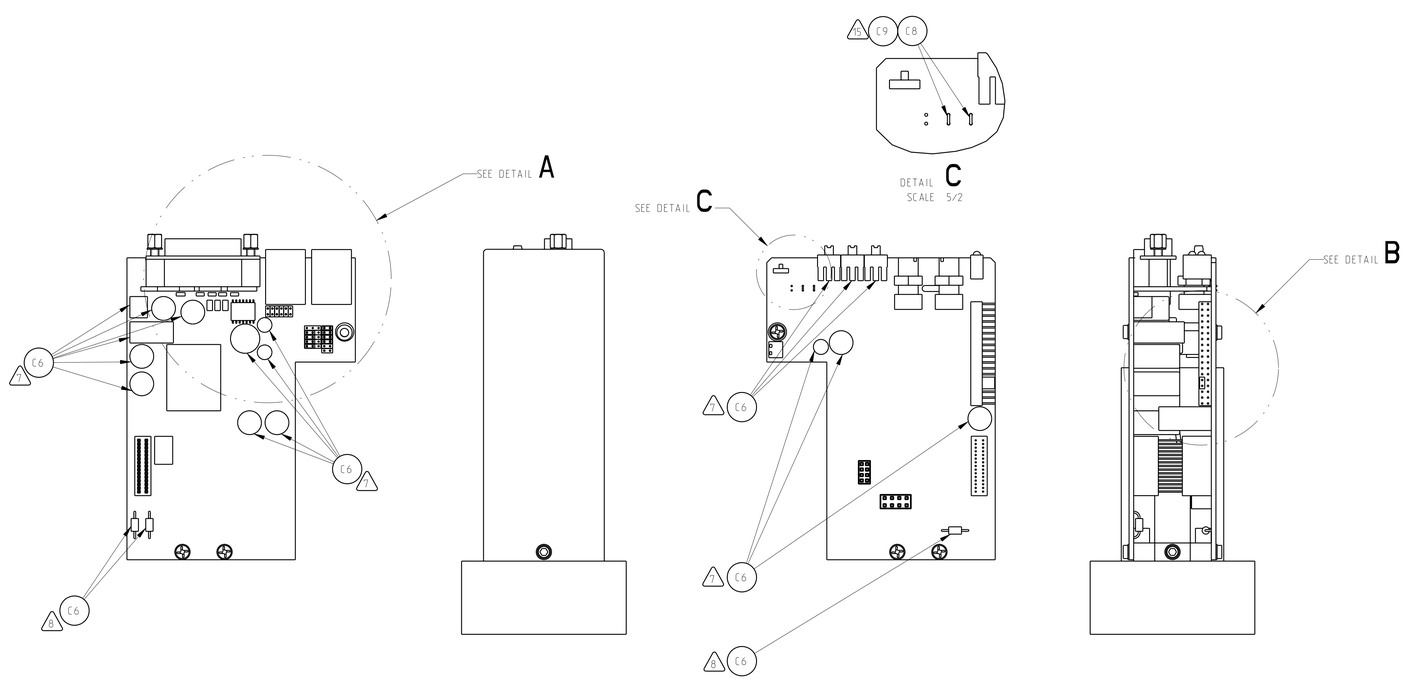
COMMERCIAL PART LIST			
ITEM	QTY	MIL SPEC/VENDOR PART NUMBER	DESCRIPTION
C1	1		SEE TABLE C1
C2	10	NAS620C4L	FLAT WASHER, REDUCED OUTSIDE DIAMETER NO. 4 (.1120), .016 THICK, CRES
C3	6	NAS1101E04-4	SCREW, MACHINE-FLAT FILLISTER HEAD .112-40 UNJC-3A X .250 LENGTH
C4	2	NAS1352N04-4	SCREW, CAP, SOCKET HEAD, A286, PASSIVATED .1120-40 UNRC-3A X .250
C5	2	JAX102-M09-F09-440	JACK SCREW, .112-40 UNC-2A X .250 LG, THREAD, 303 STAINLESS STEEL, PASSIVATED, UNICORP INC.
C6	A/R	3145 GRAY	RTV ADHESIVE/SEALANT, GRAY, DOW CORNING
C7	A/R	222MS	THREADLOCKER, LOCTITE
C8	A/R	8023 000100	BUS WIRE, 26 AWG TINNED COPPER, BELDEN
C9	A/R	SN63PB37/RMA	SOLDER, TIN/LEAD, RMA FLUX CORE

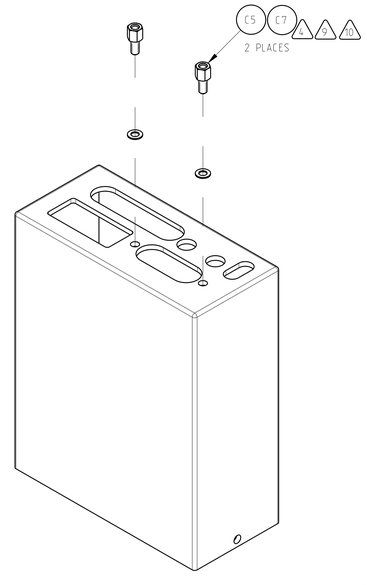
TABLE C1, PART RELATIONS		
ASSEMBLY PART NO.	VENDOR_PART_NUMBER	DESCRIPTION
S1101MFA1140-001	HFC-D-302 0-0.01SLM	MODIFIED HFC-D-302 0-0.01 SLM MASS FLOW CONTROLLER (TELEDYNE HASTINGS)
S1101MFA1140-002	HFC-D-302 0-0.02SLM	MODIFIED HFC-D-302 0-0.02 SLM MASS FLOW CONTROLLER (TELEDYNE HASTINGS)
S1101MFA1140-003	HFC-D-302 0-0.05SLM	MODIFIED HFC-D-302 0-0.05 SLM MASS FLOW CONTROLLER (TELEDYNE HASTINGS)
S1101MFA1140-004	HFC-D-302 0-0.1SLM	MODIFIED HFC-D-302 0-0.1 SLM MASS FLOW CONTROLLER (TELEDYNE HASTINGS)
S1101MFA1140-005	HFC-D-302 0-0.2SLM	MODIFIED HFC-D-302 0-0.2 SLM MASS FLOW CONTROLLER (TELEDYNE HASTINGS)
S1101MFA1140-006	HFC-D-302 0-0.5SLM	MODIFIED HFC-D-302 0-0.5 SLM MASS FLOW CONTROLLER (TELEDYNE HASTINGS)
S1101MFA1140-007	HFC-D-302 0-1SLM	MODIFIED HFC-D-302 0-1 SLM MASS FLOW CONTROLLER (TELEDYNE HASTINGS)
S1101MFA1140-008	HFC-D-302 0-2SLM	MODIFIED HFC-D-302 0-2 SLM MASS FLOW CONTROLLER (TELEDYNE HASTINGS)
S1101MFA1140-009	HFC-D-302 0-5SLM	MODIFIED HFC-D-302 0-5 SLM MASS FLOW CONTROLLER (TELEDYNE HASTINGS)
S1101MFA1140-010	HFC-D-302 0-20SLM	MODIFIED HFC-D-302 0-20 SLM MASS FLOW CONTROLLER (TELEDYNE HASTINGS)

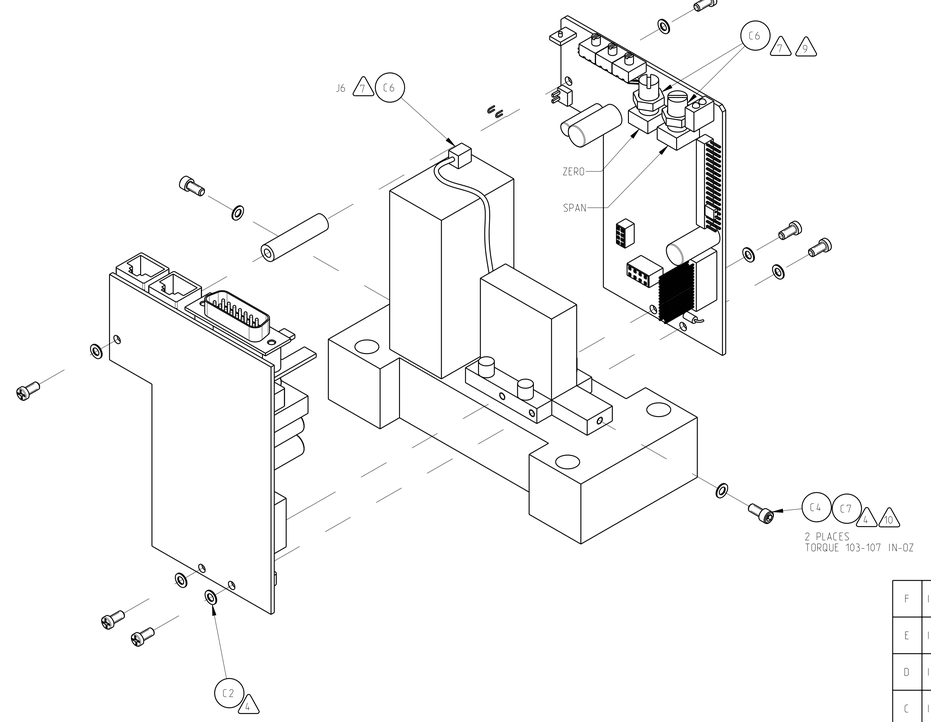
NOTES:

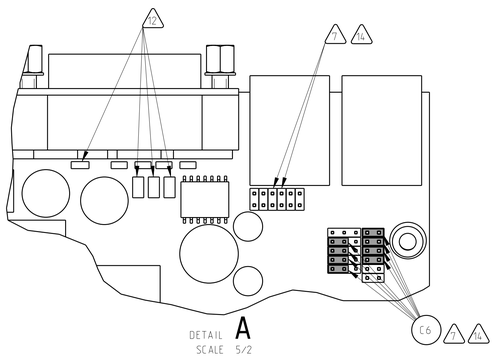
1. CERTIFICATE OF CONFORMANCE REQUIRED FOR ALL COMMERCIAL PARTS.
2. ESD HANDLING AND CONTROL PER ESTABLISHED PROCEDURES.
3. REMOVE ALL SELF-ADHESIVE STICKERS APPLIED BY TELEDYNE HASTINGS TO OUTSIDE OF MASS FLOW CONTROLLER.
4. REMOVE AND DISCARD EXISTING JACK SCREWS, PHILIPS HEAD SCREWS, AND LOCK WASHERS. REPLACE WITH ITEMS SHOWN.
5. CLEAN ALL EXTERIOR SURFACES WITH ISOPROPYL ALCOHOL AND LET AIR DRY. WIPE WITH CLEAN COTTON SWABS AND/OR LINT-FREE CLOTH.
6. HANDLE ALL PARTS AND ASSEMBLIES WITH CLEAN, LINT-FREE, ANTI-STATIC GLOVES OR FINGER COTS.
7. STAKE COMPONENTS AT DESIGNATED AREAS SHOWN. STAKING AND INSPECTION SHALL BE IN ACCORDANCE WITH NASA-STD-8739.1.
8. ENCAPSULATE GLASS DIODES WITH ITEM C6 PER ESTABLISHED PROCEDURES.
9. TIGHTEN FASTENERS SNUG TIGHT PER ESTABLISHED PROCEDURES
10. APPLY THREAD LOCKING COMPOUND PER ESTABLISHED PROCEDURES.
11. BEFORE STORAGE, PROTECT WITH CONTAMINATION BARRIER. USE CLEAN, NEW, ESD BAG, TOTES OR APPROVED ESD PACKAGING. MARK PART NUMBER, A DASH, AND A SERIAL/LOT NUMBER AS SPECIFIED IN WORK ORDER/SOW. BY TAGGING PER SAE-AS478-35D.
12. IF PRESENT, REMOVE AND DISCARD RESISTORS R476, R477, R478 AND R479.
13. TEST MASS FLOW CONTROLLER C1 BEFORE AND AFTER MODIFICATION PER ACME-TST-030.
14. CONFIGURE JUMPERS AS SHOWN. DISCARD ANY UNNEEDED JUMPERS.
15. REMOVE DIP SWITCH AND ADD BUS WIRE, ITEM C8, USING SOLDER, ITEM C9, AS SHOWN.



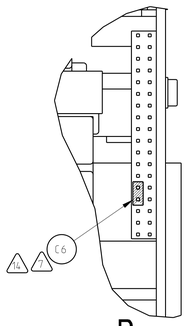
DETAIL C
SCALE 5/2



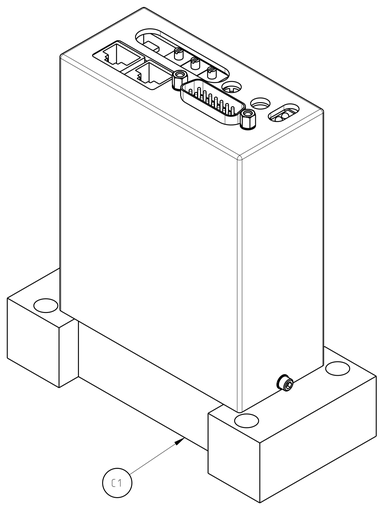




DETAIL A
SCALE 5/2



DETAIL B
SCALE 5/2



S1101MFA1140-001 THRU -010

REV	DESCRIPTION	APPROVED
F	INCORPORATED ACME-ECO-505	JO 09/24/15 WB 09/25/15 JB 10/14/15
E	INCORPORATED ACME-ECO-437	JO 04/30/15 GF 05/01/15 AD 05/08/15
D	INCORPORATED ACME-ECO-394	KR 11/18/14 AD 11/21/14 GF 11/20/14
C	INCORPORATED ACME-ECO-377	KR 10/23/14 AD 10/27/14 GF 10/29/14
B	INCORPORATED ACME-ECO-285	KR 04/30/14 GF 05/05/14 AD 05/01/14
A	INCORPORATED ACME-ECO-262	MA 01/23/14 GF 01/23/14 AD 01/24/14

REVISION HISTORY

NEXT ASSEMBLY

UNLESS OTHERWISE SPECIFIED:
1. ALL DIMENSIONS ARE IN INCHES
2. BREAK EDGES .010-.020
3. INSIDE CORNERS R.02 MAX
4. LINEAR TOLERANCES:
   .X ± .03
   .XX ± .01
   .XXX ± .005
5. ANGULAR TOLERANCE ± .5°

THIRD ANGLE PROJECTION

THIS DRAWING IS UNDER CONFIGURATION CONTROL. DO NOT REVISE WITHOUT AN APPROVED ECO.
APPROVED: S.COBB-CAIN 10/15/15
PRIOR TO USE, CHECK WITH CONFIGURATION MANAGEMENT FOR ANY OUTSTANDING ECOS.

DESIGNER	R. REID 11/05/13	QUALITY	M. BORRELLI 11/06/13
REVIEWER	J. OGRIN 11/13/13	MATERIALS	M. WILKINSON 11/06/13
MECHANICAL	A. DRAKE 11/06/13	PROJ LEAD	B. BOROWSKI 11/15/13
ELECTRICAL	T. GOBLE 11/12/13		
STRUCTURAL	C. KERNER 11/14/13		
SAFETY	D. SEELEY 11/07/13		

DIMENSIONS AND TOLERANCING PER ASME Y14.5M-1994

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www.zin-tech.com

CONTRACT NUMBER: NNC09BA02B
PROJECT CODE: SPACE 110

NATIONAL AERONAUTICS AND SPACE ADMINISTRATION
JOHN H. GLENN RESEARCH CENTER
LEWIS FIELD
CLEVELAND, OHIO

TITLE: MODIFIED MASS FLOW CONTROLLER

SIZE	CAGE CODE	DWG NO	REV
E	0SG68	S1101MFA1140	F

SCALE 5/4 | INITIAL REL DATE 11/15/2013 | SHEET 1 OF 1

RELEASE LEVEL: RELEASED
MODEL: S1101MFA1140

PRO/E VERSION: Wildfire 5.0

DWG NO S1101MFA1140 REV F SH 1